Bahria University, Islamabad

Department of Software Engineering

Cloud Computing Lab (Spring-2026)

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Task No:	Task Wise Marks		Documentation Marks		Total Marks (20)
	Assigned	Obtained	Assigned	Obtained	
1	3		5		
2	3				
3	3				
4	3				
5	3				

Comments:

Signature

Lab No: 02 – Docker Guide

Introduction

Introduction to Container and Images in Docker.

Tools Used

- Docker

Docker Pull:

```
bash-5.2$ docker pull bash
Using default tag: latest
latest: Pulling from library/bash
d8ad8cd72600: Already exists
44c05f528325: Pull complete
8d3ad8d18022: Pull complete
Digest: sha256:e320b40b14abc61f053286705070342c46034a549a276b798e64541457c4af92
Status: Downloaded newer image for bash:latest
docker.io/library/bash:latest

What's next:
    View a summary of image vulnerabilities and recommendations → docker scout quickview bash
bash-5.2$
```

Figure 1. Pulling Bash Image from Docker Hub

```
bash-5.2$ docker pull gradle
Using default tag: latest
latest: Pulling from library/gradle
66a4bbbfab88: Already exists
eebd9d6193ad: Pull complete
fc9c2f0cb67d: Pull complete
4f4fb700ef54: Pull complete
b8d7d995e437: Pull complete
05f3bb66fe06: Pull complete
edd6674e7bef: Pull complete
c5dba3a14ccc: Pull complete
b29991c76749: Pull complete
Digest: sha256:85aec999629f4774a383cb792da4b598bdf5a7e69c4b9570bb70c0f919179183
Status: Downloaded newer image for gradle:latest
docker.io/library/gradle:latest

What's next:
    View a summary of image vulnerabilities and recommendations → docker scout quickview gradle
bash-5.2$
```

Figure 2. Pulling Gradle Image from Docker Hub

Docker Run:

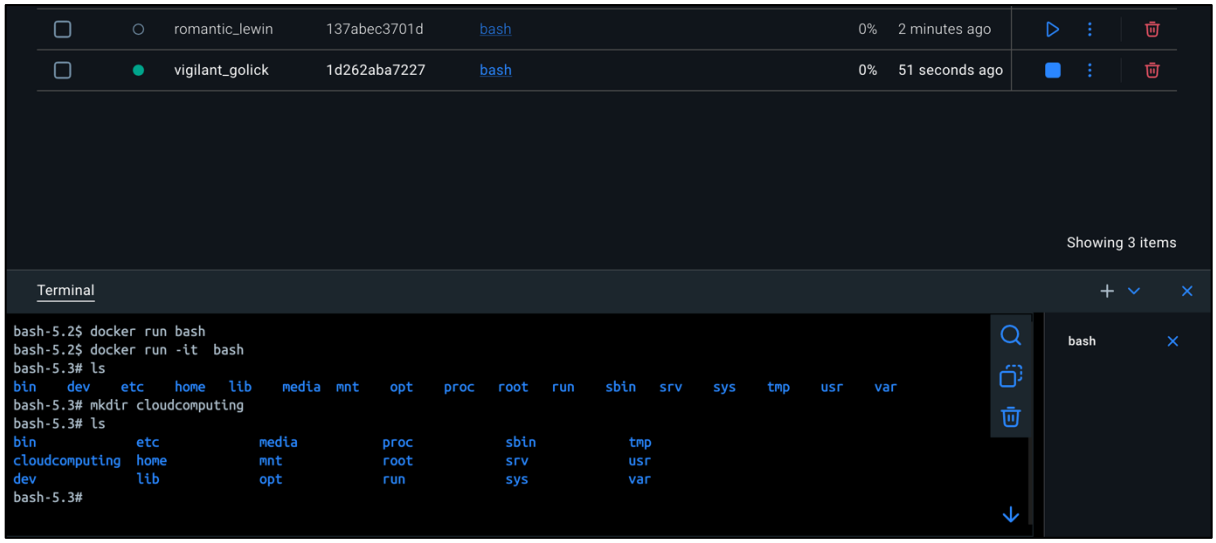

Figure 3. Running Bash Image

Docker Exit:

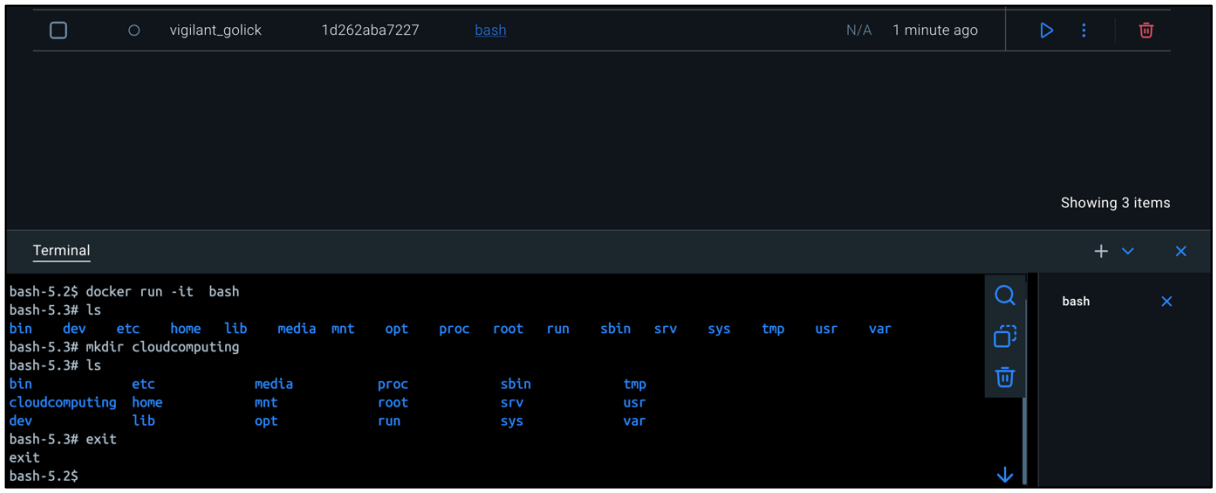

Figure 4. Exiting the Container

Docker ps -a & ps:

```
bash-5.2$ docker ps -a
CONTAINER ID   IMAGE     COMMAND                  CREATED              STATUS                     PORTS     NAMES
1d262aba7227   bash      "docker-entrypoint.s…"   About a minute ago   Exited (0) 28 seconds ago            vigilant_golick
137abec3701d   bash      "docker-entrypoint.s…"   2 minutes ago        Exited (0) 2 minutes ago             romantic_lewin
9b101f556497   ubuntu    "/bin/bash"              31 minutes ago       Exited (0) 27 minutes ago            festive_keller
bash-5.2$ docker ps
CONTAINER ID   IMAGE     COMMAND   CREATED   STATUS    PORTS     NAMES
bash-5.2$
```

Figure 5. Docker ps commands

Docker Images:

```
bash-5.2$ docker images
                                                                    i Info →   U  In Use
IMAGE           ID             DISK USAGE   CONTENT SIZE   EXTRA
bash:latest     7de897b5d1c5       18.2MB             0B    U
gradle:latest   d3d23d6a4fbf        779MB             0B
ubuntu:latest   350d40843c24        101MB             0B    U
bash-5.2$
```

Figure 6. All of the Docker Images

Docker Start and Stop:

```
bash-5.2$ docker start 1d262aba722756b0dd14a577d7ed9e6cf981133cca5d65078853676595aa763f
1d262aba722756b0dd14a577d7ed9e6cf981133cca5d65078853676595aa763f
bash-5.2$ docker stop 1d262aba722756b0dd14a577d7ed9e6cf981133cca5d65078853676595aa763f
1d262aba722756b0dd14a577d7ed9e6cf981133cca5d65078853676595aa763f
bash-5.2$
```

Figure 7. Start and Stop Command

Docker Delete Commands:

```
bash-5.2$ docker rm festive_keller
festive_keller
bash-5.2$ docker rmi ubuntu
Untagged: ubuntu:latest
Untagged: ubuntu@sha256:d1e2e92c075e5ca139d51a140fff46f84315c0fdce203eab2807c7e495eff4f9
Deleted: sha256:350d40843c2436d6203ee1e35e9264fb08194c3455ec7462236e01adf542c00b
bash-5.2$
```

Figure 8. Docker rmi and rm commands